Day 9 — Data Preparation & Split Strategy

Dataset & Annotation Findings

Dataset Findings

Dataset	Findings
DIATAquarium	11 classes; uneven class distribution; local test set not available. data.yaml points both val and test to test/images.
Aquatic Plant	1 class; 13 training images with no matching label; 6 duplicate images found in the training sample.
Well	4 classes; severe class imbalance — fishes: 50,194, Inlet-pipe: 3, stone: 5.

Representative Difficult Samples

1. Low-contrast / low-visibility example

2. Background image with empty annotation

3. Object detection example

Image Quality & Resolution

- DIATAquarium train images: 1920 × 1080
- DIATAquarium validation images: 2048 × 1536
- Aquatic Plant: 640 × 640
- Well: 640 × 640
- Some sampled images showed haze, low contrast, reduced visibility and colour variation.

Annotation Findings

- YOLO annotation format was checked for all 3 datasets.
- 0 invalid annotation entries were found.
- Empty labels may represent valid background/negative images.
- Sampled bounding boxes generally matched the visible objects.

Confirmed Data Preparation Strategy

Preparation Pipeline

Raw Dataset → Quality Check → Annotation Review → Preprocessing → Train / Val / Test → YOLO Training

Preprocessing Plan

- Preserve the original/raw datasets.
- Do not permanently modify raw data.
- Standardize image size before model training.
- Use White Balance + CLAHE as the classical enhancement method.
- Keep original images as the baseline.
- Review images without matching labels.
- Retain valid empty / background samples.

Important Note

The data-preparation strategy is being frozen on Day 9; the dataset itself will be checked further during preprocessing before Dataset V1 is frozen.

Split Strategy

- Train → model learning
- Validation → tuning and model selection
- Test → final evaluation when available
- Preserve existing valid splits.
- Do not mix images between train, validation and test.

DIATAquarium Split Issue

The local data.yaml points both val and test to test/images, while no local test folder exists. Final handling will follow the mentor-approved strategy before training.

Frozen Data-Preparation Plan

Dataset Scope

All three supplied datasets will be used separately.

Baseline

Original Images → YOLO Object Detection

Enhanced Experiment

Original Images → White Balance + CLAHE → YOLO Object Detection

Advanced Enhancement

Water-Net / FUnIE-GAN will be considered as learning-based enhancement methods, per the earlier experiment plan and implementation feasibility.

Metrics (from modelling stage onward)

- Precision
- Recall
- mAP@0.5
- mAP@0.5:0.95

Data Quality Rules

- Keep the raw datasets unchanged.
- Review missing-label images.
- Retain valid empty / background samples.
- Review duplicate and near-duplicate images.
- Check for possible data leakage before final dataset freezing.
- Correct only confirmed annotation errors.
- Standardize image size.
- Maintain train / validation / test separation.